

Topic 7: Intermolecular Forces

This comprehensive guide covers every specification point in **Topic 7 (Intermolecular Forces)** for Edexcel IAL Chemistry Unit 2, detailed step by step 1.

Specification Point 7.1: Understand the Nature of Intermolecular Forces

Intermolecular forces (non-bonded interactions) are weak attractive forces that exist **between separate molecules or uncombined noble gas atoms**, distinct from strong intramolecular ionic, metallic, or covalent bonds within molecules 2-5.

7.1(i) London Forces (Instantaneous Dipole–Induced Dipole / Dispersion Forces)

- **Mechanism:** Electrons within an atom or non-polar molecule are in continuous, random motion 6-8. At any given instant, the electron cloud density may fluctuate unevenly across the molecule, giving rise to an **instantaneous (temporary) dipole** 6-8.
- **Induction:** This temporary dipole distorts the electron cloud of a neighbouring molecule, inducing an **induced dipole** of opposite charge 6-8.
- **Attraction:** The δ^+ end of one temporary dipole attracts the δ^- end of the neighbouring induced dipole 6, 9, 10.
- **Occurrence:** London forces exist between **all atoms and molecular species** (whether polar or non-polar), as well as monatomic noble gases, but do not exist in giant ionic lattices 9, 11, 12.

7.1(ii) Permanent Dipole–Permanent Dipole Interactions

- **Mechanism:** These forces occur between **polar molecules** that possess permanent dipoles resulting from differences in electronegativity across covalent bonds 13-16.
- **Attraction:** Electrostatic attraction occurs between the permanent δ^+ region of one molecule and the permanent δ^- region of an adjacent molecule 14-16.
- **Relative Strength:** For small molecules containing similar numbers of total electrons, permanent dipole–dipole interactions are generally **stronger than London forces** 13, 15, 17. For instance, propanone (polar) has a higher boiling temperature (56°C) than butane (non-polar, 0°C) because more energy is required to break permanent dipole forces 13, 17.

7.1(iii) Hydrogen Bonds

1. **Mechanism:** Hydrogen bonding is a particularly strong, specialized form of permanent dipole–permanent dipole interaction 3, 7, 18, 19.
2. **Requirements:**
3. A hydrogen atom covalently attached directly to one of the **three most electronegative elements: Nitrogen (N), Oxygen (O), or Fluorine (F)** 18-20.
4. A **lone pair of electrons** on the N, O, or F atom of an adjacent molecule 18-20.
5. **Explanation:** The high electronegativity difference strongly pulls shared electrons away from the hydrogen atom 18, 19, 21. Because hydrogen has no inner shell of shielding electrons, its positive nucleus is exposed, creating a high δ^+

charge density that strongly attracts the lone pair on a neighbouring N, O, or F atom 18, 19, 21.

6. **Relative Strength Hierarchy: Covalent bonds $\gg$ Hydrogen bonds $>$ Permanent dipole–dipole forces $>$ London dispersion forces** 5, 7, 11.

Specification Point 7.2: Hydrogen Bonding in H_2O , Liquid NH_3 , and Liquid HF

Hydrogen bonding significantly dictates the behaviour of simple hydrides containing N, O, and F 18, 20, 22:

- **Water (H_2O):**
 - Each oxygen atom has **two O–H bonds and two lone pairs of electrons** 22, 23.
 - Water molecules can form an extensive 3D hydrogen-bonded network with up to **four hydrogen bonds per molecule** 22-24.
- **Ammonia (NH_3):**
 - Nitrogen forms three N–H bonds and possesses **one lone pair of electrons** 25.
 - Because it has only one lone pair, ammonia forms an average of **one hydrogen bond per molecule**, making its intermolecular attraction less extensive than water's 22, 25.
- **Hydrogen Fluoride (HF):**
 - Fluorine is the most electronegative element and has **three lone pairs**, but only **one hydrogen atom per molecule** 25, 26.
 - HF forms long linear hydrogen-bonded chains $(\text{HF})_n$ 25. Despite having strong individual hydrogen bonds, HF forms fewer hydrogen bonds per molecule overall compared to water 22, 24.

Specification Point 7.3: Anomalous Properties of Water Resulting from Hydrogen Bonding

Water exhibits two major anomalous physical properties directly caused by its extensive hydrogen bonding network 22, 27, 28:

7.3(i) High Melting and Boiling Temperatures

- **Observation:** Water has anomalously high melting and boiling points (and enthalpy of vaporisation) compared to other Group 6 hydrides (H_2S , H_2Se , H_2Te) and molecules of comparable molar mass 22, 27-30.
- **Explanation:** While H_2S , H_2Se , and H_2Te are held together primarily by weaker London dispersion forces, liquid water requires a large amount of thermal energy to overcome its extensive network of strong hydrogen bonds 22, 29, 30.

7.3(ii) Lower Density of Ice Compared to Liquid Water

- **Observation:** Ice floats on liquid water because solid ice at 0°C is **less dense** than liquid water at 0°C 22, 28.
- **Explanation:** In liquid water, hydrogen bonds constantly break and re-form dynamically 31. When water freezes into ice, the molecules are locked into a fixed, rigid 3D hexagonal ring structure held open by 4 hydrogen bonds per molecule 22,

31, 32. This arrangement creates an **open, cage-like structure with large empty spaces** 22, 31, 32. When ice melts, this open ring framework collapses, allowing water molecules to pack closer together, which increases the liquid's density 22, 31, 32.

Specification Point 7.4: Predicting Hydrogen Bonding in Analogous Molecules

Hydrogen bonding can be predicted in any compound where hydrogen is bonded to nitrogen or oxygen 21, 22, 33:

- **Alcohols** (R-OH): Form intermolecular hydrogen bonds via the hydroxyl group 21, 22, 33.
- **Carboxylic Acids** (R-COOH): Form hydrogen-bonded dimers in non-polar solvents 25, 34.
- **Amines** (R-NH_2), **Amides** (R-CONH_2): Form hydrogen bonds via N-H groups and lone pairs on nitrogen 22.
- **Proteins & Biological Macromolecules**: Rely on $\text{N-H}\cdots\text{O}=\text{C}$ hydrogen bonds to maintain secondary structures 22.

Specification Point 7.5: Intermolecular Forces and Physical Properties

7.5(i) Boiling Temperatures of Alkanes with Increasing Chain Length

- **Trend**: Boiling temperatures of unbranched alkanes increase steadily as carbon chain length increases 9, 35.
- **Explanation**: As chain length grows, relative molecular mass and the total number of electrons per molecule increase 9, 35, 36. Larger electron clouds undergo greater fluctuations in electron density, generating **stronger instantaneous and induced dipoles** 9, 35, 36. Furthermore, longer straight chains offer a **larger surface area of contact** between molecules, increasing total London dispersion forces and requiring more energy to separate molecules 15, 35, 36.

7.5(ii) Effect of Branching on Alkane Boiling Temperatures

- **Trend**: Branched-chain alkanes have **lower boiling temperatures** than their unbranched structural isomers 15, 37-39.
- **Explanation**: Branching makes molecules more compact and spherical, which **reduces the surface area of contact** between adjacent molecules 15, 37-39. Fewer points of contact weaken the overall London dispersion forces, requiring less thermal energy to overcome 15, 37-39. For example, pentane boils at 309 K , 2-methylbutane at 301 K , and 2,2-dimethylpropane at 283 K 39.

7.5(iii) Low Volatility (Higher Boiling Points) of Alcohols Compared to Alkanes

- **Trend**: Alcohols have significantly higher boiling points (lower volatility) than alkanes with similar electron numbers or chain lengths 22, 39, 40.
- **Explanation**: Alkanes are non-polar and held together solely by weak London dispersion forces 22, 39, 40. Alcohols contain the polar -OH group, allowing them to

form **strong intermolecular hydrogen bonds in addition to London forces** 22, 39, 40. Breaking these extra hydrogen bonds requires substantially higher thermal energy 22, 39, 40. For instance, propane boils at -42°C while propan-1-ol boils at 97°C 40, 41.

7.5(iv) Trends in Boiling Temperatures of Hydrogen Halides (HF to HI)

1. **Observation:** HF has an anomalously high boiling point (293°K) 42-44. From HCl to HI , boiling points drop sharply to HCl (188°K) and then rise steadily through HBr (207°K) to HI (238°K) 42-44.
2. **Explanation:**
3. **HF Anomaly:** HF forms strong intermolecular **hydrogen bonds** due to the extreme polarity of the H-F bond and small, intense lone pairs on fluorine 26, 42-44.
4. **HCl to HI Trend:** HCl , HBr , and HI cannot form hydrogen bonds; they experience permanent dipole-dipole forces and London forces 42, 43, 45. Going from HCl to HI , electronegativity decreases, so permanent dipole forces weaken 28, 43. However, the **number of electrons increases significantly** ($\text{HCl} < \text{HBr} < \text{HI}$), strengthening the **London dispersion forces** enough to overcome the drop in permanent dipole attractions and increase the boiling point 28, 42, 43.

Specification Point 7.6: Factors Influencing Solvent Choice

Dissolution depends on the balance of intermolecular interactions: a solute dissolves when the **solute-solvent attractions are strong enough to overcome both solute-solute and solvent-solvent forces** ("like dissolves like") 26, 43, 46, 47.

7.6(i) Water as a Solvent for Ionic Compounds

- **Mechanism:** Giant ionic lattices are held together by strong electrostatic ionic bonds 46.
- **Hydration:** Water is a polar solvent ($\delta^+ \text{H}$, $\delta^- \text{O}$) 43, 46. When an ionic solid is placed in water, the δ^+ hydrogen atoms attract negative anions, while δ^- oxygen atoms attract positive cations 43, 48.
- **Energetics:** The energy released during ion hydration (**hydration enthalpy**) supplies the energy needed to break apart the ionic lattice (**lattice enthalpy**) 43, 46.

7.6(ii) Water as a Solvent for Simple Alcohols

- **Mechanism:** Small alcohols (methanol, ethanol, propan-1-ol) are completely soluble in water because their hydroxyl ($-\text{OH}$) groups form **hydrogen bonds with water molecules** 33, 43, 48.
- **Chain Length Effect:** As the non-polar hydrocarbon chain lengthens (e.g., hexanol, octanol), solubility in water decreases markedly 33, 43, 48, 49. The non-polar alkyl chain cannot form hydrogen bonds with water and disrupts water's hydrogen-bonded network; hydrophobic London forces between long alkyl chains become predominant 33, 43, 49.

7.6(iii) Water as a Poor Solvent for Halogenoalkanes and Non-Polar Compounds

- **Explanation:** Halogenoalkanes contain polar C--X bonds but **lack H--F, H--O, or H--N bonds**, meaning they **cannot form hydrogen bonds with water** 43, 50, 51.
- **Energetics:** The permanent dipole–dipole forces formed between halogenoalkanes and water are too weak to break the strong hydrogen bonds between water molecules, rendering halogenoalkanes insoluble or only sparingly soluble in water 43, 50, 51.

7.6(iv) Non-Aqueous Solvents for Non-Polar Compounds

- **Explanation:** Non-polar solutes (such as alkanes or iodine) readily dissolve in **non-polar organic solvents** (such as hexane or cyclohexane) 26, 43, 47, 52.
- **Interactions:** The weak London forces broken between solute molecules and between solvent molecules are replaced by London forces of comparable strength between solute and solvent molecules 43, 47, 52.

 **Next Step:** Would you like to practice exam-style questions on Topic 7, or move on to **Topic 8 (Redox Chemistry and Groups 1, 2, and 7)** next?